Gradient Boosting:

This model instead of generating the output it generates the mena followed by Error and based on same it further creates the predicted y values

It sequentially trains models to correct the mistakes of previous models by focusing on the **residual errors.**

At every stage using DT it generates the Errors then it combines the Error that stage with the mean predicted by 1st DT. Repeats the process until max accuracy

It is effective at capturing complex patterns and is widely used for regression, classification, and ranking tasks.

1st Stage: Decision Tree predicts the mean of the target.
2nd Stage: A new Decision Tree learns the errors/residuals of the first model.
3rd Stage: Another Decision Tree learns the remaining errors after the second model.
Repeat: Each new tree focuses on reducing the remaining error, and its prediction is added to the previous prediction.

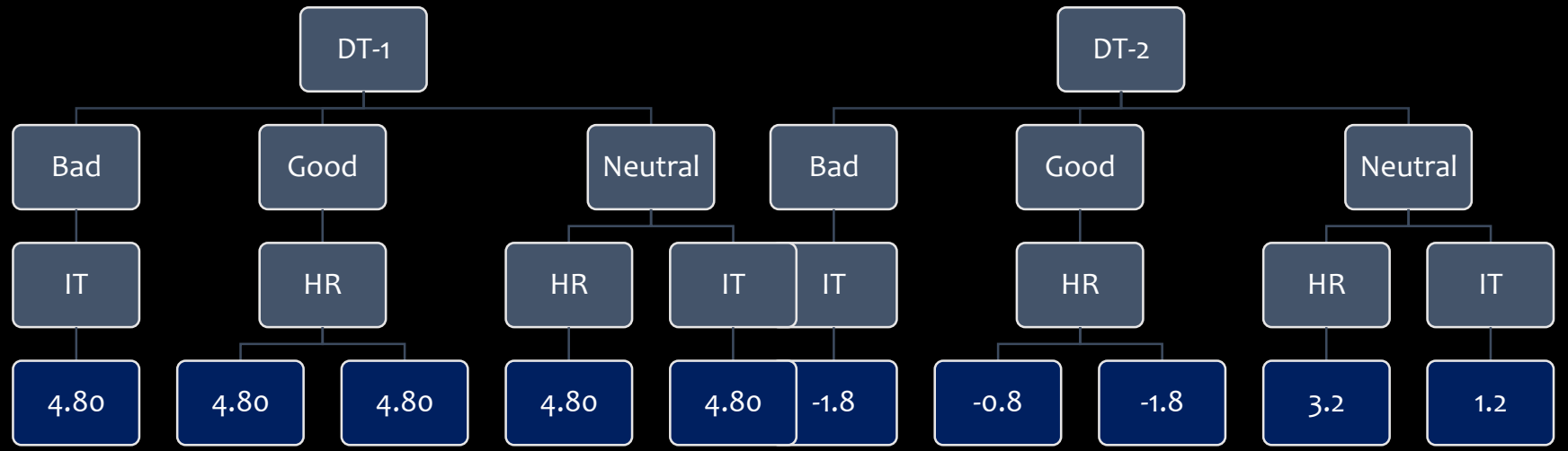
Sr No	Department	Credit	Salary (Actual_y)	Mean	Residual _M2	m2	
						Without Lr	With Lr
						y_predict Salary	Calculate Salary
1	IT	Bad	3	4.80	-1.80	3.00	4.26
2	HR	Good	4	4.80	-0.80	4.00	4.56
3	HR	Neutral	8	4.80	3.20	8.00	5.76
4	IT	Neutral	6	4.80	1.20	6.00	5.16
4	HR	Good	3	4.80	-1.80	3.00	4.26
Total							

$$Prediction = m_1 + Lr(Err\ of\ m_2)$$

$$Prediction = 4.80 + 0.3(-1.80)$$

$$Prediction = 4.80 + (-0.54)$$

$$Prediction = 4.26$$



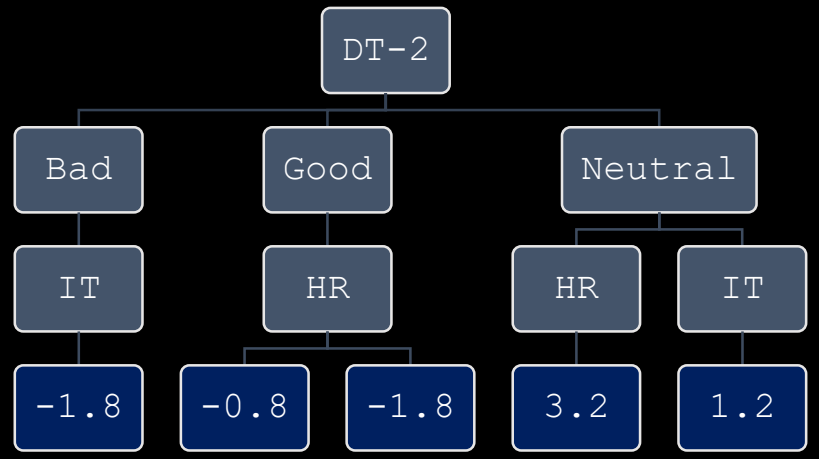
	m1	m2	
			With Lr
Salary (Actual_y)	Mean	Residual _M2	Calculate Salary
3	4.80	-1.80	4.26
4	4.80	-0.80	4.56
8	4.80	3.20	5.76
6	4.80	1.20	5.16
3	4.80	-1.80	4.26
Total			

$$Prediction = m_1 * Lr(Err\ of\ m_2)$$

$$Prediction = 4.80 * 0.3(-1.80)$$

$$Prediction = 4.80 + (-0.54)$$

$$Prediction = 4.26$$



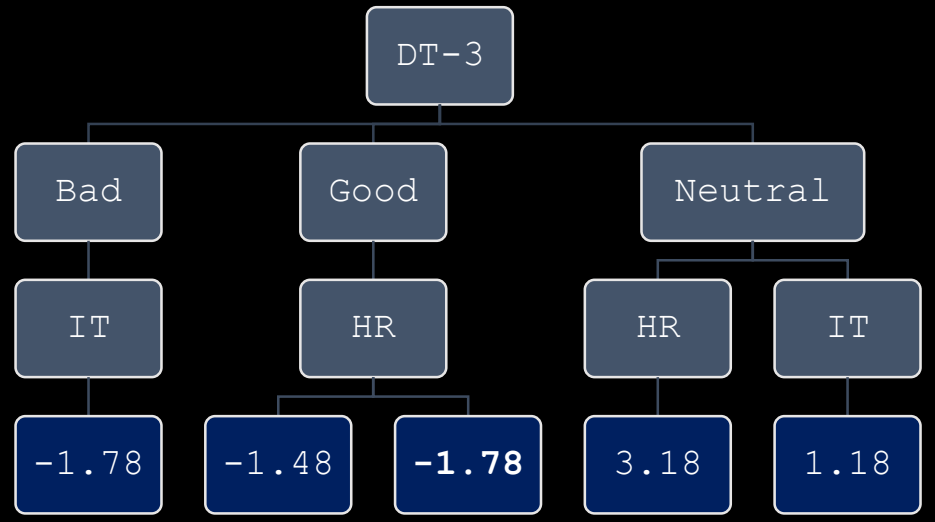
	m1	m2		m3	
			With Lr		With Lr
Salary (Actual_y)	Mean	Residual _M2	Calculate Salary	Residual M3	Calculate Salary
3	4.80	-1.80	4.26	-1.78	3.73
4	4.80	-0.80	4.56	-1.48	4.12
8	4.80	3.20	5.76	3.18	6.71
6	4.80	1.20	5.16	1.18	5.51
3	4.80	-1.80	4.26	-1.78	3.73
Total					

$$Prediction = m_1 + Lr(Err\ of\ m_2) + Lr(Err\ of\ m_3)$$

$$Prediction = 4.80 + 0.3(-1.80) + 0.3(-1.78)$$

$$Prediction = 4.80 + (-0.54) + (-0.53)$$

$$Prediction = 3.73$$



	m1	m2		m3		m4	
			With Lr		With Lr		With Lr
Salary (Actual_y)	Mean	Residual _M2	Calculate Salary	Residual M3	Calculate Salary	Residual M4	Calculate Salary
3	4.80	-1.80	4.26	-1.78	3.73	-1.48	3.28
4	4.80	-0.80	4.56	-1.48	4.12	-1.18	3.76
8	4.80	3.20	5.76	3.18	6.71	2.88	7.58
6	4.80	1.20	5.16	1.18	5.51	0.88	5.78
3	4.80	-1.80	4.26	-1.78	3.73	-1.48	3.28
Total							

$$Prediction = m_1 + Lr(Err\ of\ m_2) + Lr(Err\ of\ m_3) + Lr(Err\ of\ m_4)$$

$$Prediction = 4.80 + 0.3(-1.80) + 0.3(-1.78) + 0.3(-1.48)$$

$$Prediction = 4.80 + (-0.54) + (-0.53) + (-0.44)$$

$$Prediction = 3.28$$

